Design and Simulation-Based Evaluation of a Three-Aircraft Autonomous System for Post-Flood Search, Geolocation and Relief Delivery

Swastik Kumar
Department of Electronics & Communication Engineering
Thapar Institute of Engineering & Technology
Patiala, Punjab, India
skumar6_be24@thapar.edu

***Abstract*—Finding survivors is usually the slowest step of a flood rescue. We describe RescueSwarm, a system of three multirotor aircraft that search an assigned area under one operator, detect people from 40 m, report each one as a map coordinate and drop a 200 g relief kit to them, all without a mobile network. The paper covers the design and its evaluation by analysis and software-in-the-loop simulation; no aircraft has flown yet. Four results stand out. First, a Monte Carlo geolocation budget shows that about 0.88 m of coordinate error comes from sources no receiver can remove, so any receiver with horizontal error below 2.75 m meets a 5 m delivery requirement at 95 % confidence; RTK gives 1.53 m and standalone NavIC, at 8.79 m, does not. Second, a person in water at 40 m covers 1 498 px^2 only if the 12 MP frame is tiled at full resolution, and either halving the resolution or flying at 60 m pushes the target into the small-object range where detectors are weakest. Third, a single 802.11 mesh would have left 8.7 dB of margin at the 600 m geofence, so we split traffic over three radios with at least 29.7 dB each, and found that LoRa framing overhead, not bit rate, limits the coordination channel. Fourth, in simulation with three ArduCopter autopilots, staggering take-off in the mission file raised the closest approach at launch from 1.31 m to 26.96 m.**

***Index Terms*—unmanned aerial systems, multi-robot coordination, search and rescue, geolocation error, small object detection, link budget, disaster response**

I. Introduction

India recorded 5.4 million internal displacements from disasters in 2024, the most in twelve years, and 2.4 million of them came from one monsoon flood season in Assam [1]. In a flood, boat crews and ground teams reach people reliably once they know where to go. The hard part is knowing where to go. Boats search slowly and cover rooftops and debris poorly, and the few helicopters available are mostly flying evacuations.

Small multirotors could take over much of that search, but one aircraft carrying a useful payload for fifteen minutes covers little ground, and a team that needs one pilot per aircraft does not scale. We therefore set out to design a small fleet that one person can supervise. Three aircraft divide an area, sweep it, detect people on board, agree among themselves who serves each survivor, and drop a kit. The design target was a first-response team in the field, so the system carries its own positioning base and radios and needs no internet or mobile service.

Building such a system mostly means trading one requirement against another, and the trades are where we think the useful findings are. This paper reports them. Its contributions are:

1) a geolocation error budget, checked by Monte Carlo simulation, that gives the receiver accuracy needed for a 5 m delivery requirement and shows how little of the error the receiver controls;
2) an analysis of detection geometry for a person in water, showing why survey altitude and tiling resolution have to be chosen together;
3) a link and channel-occupancy analysis that led us from one mesh network to three single-purpose radios;
4) a battery analysis covering pack selection at the aircraft's own mass and the dynamic voltage sag seen by the low-battery failsafe; and
5) software-in-the-loop results for launch, sweep and recovery with three autopilots running together.

None of the aircraft has flown. Every result here is either computed from a model or measured in simulation, and we say which in each case.

II. Related Work

Coverage and allocation. DARP divides an area into equal, connected regions, one per robot [2], and boustrophedon decomposition remains the usual way to sweep a rectangular sensor footprint across a region [3]. For assigning tasks among agents, the consensus-based bundle algorithm is the common decentralised choice [4]. We use a fixed equal-area split with boustrophedon coverage inside each strip, and a greedy claim-and-lock rule for survivors. With three aircraft, each linked to one ground station, the guarantees that CBBA offers add little, and a plan that is the same every time the inputs are the same is much easier to test.

Detecting people from the air. People seen from survey altitude are tiny. COCO counts anything under 32^2 px^2 as a small object [5], and TinyPerson adds finer bands below that because aerial images of people are often smaller still [6]. Running the detector on overlapping crops of the full-resolution frame, as in SAHI, recovers much of the recall lost on small objects [7].

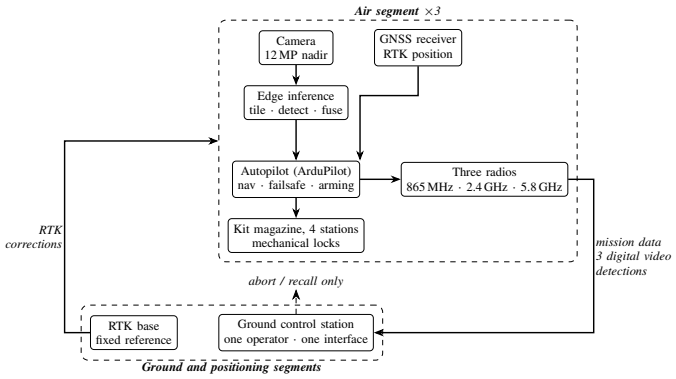


Fig. 1. System architecture. Detection runs on board, so the downlink serves the operator rather than the detector. The only commands the ground station can send are abort and recall.

Relevant datasets include SARD for land search [8], VisDrone for urban scenes [9] and SeaDronesSee, which labels people in open water [10]. The last is closest to a flood, with a low-contrast background and targets that show only head and shoulders, but none of them covers flooded Indian terrain.

Airdrop and positioning. Mathisen *et al.* reported autonomous ballistic airdrop from a small fixed-wing aircraft with a mean error of 5.5 m, and found that aircraft velocity and wind at release dominate the error [11]. A multirotor can almost stop before releasing, which removes the largest of those terms. For positioning without a network, real-time kinematic (RTK) correction from a local base station is the standard way to reach sub-metre relative accuracy [12].

III. System Design

A. Architecture

Fig. 1 shows the three parts of the system. Each aircraft is an 18 in quadrotor with a Holybro Kakute H7 autopilot running ArduPilot [15], a u-blox ZED-F9P dual-band RTK receiver [16], a downward-facing 12.3 MP camera, a Raspberry Pi 5 companion computer with a 26 TOPS neural accelerator, a four-station kit magazine with mechanical locks, and three radios. The ground segment is a single station and one operator. A second ZED-F9P on a tripod acts as the RTK base and sends corrections over its own 865 MHz radio. The companion forwards them to the autopilot, which passes them to the receiver.

The ground station cannot retask an aircraft, move a waypoint or order a drop. We left those commands out of the software entirely rather than disabling them, so they cannot be switched on by mistake. Safety control comes from a separate 2.4 GHz ExpressLRS link wired straight to the autopilot, so the safety pilot keeps control if the companion computer hangs.

B. Mission and coordination

During setup the operator draws the search area, which is split into three strips of equal area, one per aircraft. Each strip is swept back and forth at 40 m and 8 m/s. Before upload, the planner tries all four combinations of starting side and sweep direction and keeps the one that ends nearest the pad, because the return leg is flown on the emptiest battery. Take-off delays of 0, 15 and 30 s and transit altitudes of 20, 25 and 30 m are written into the mission files, so separation does not depend on the operator's timing.

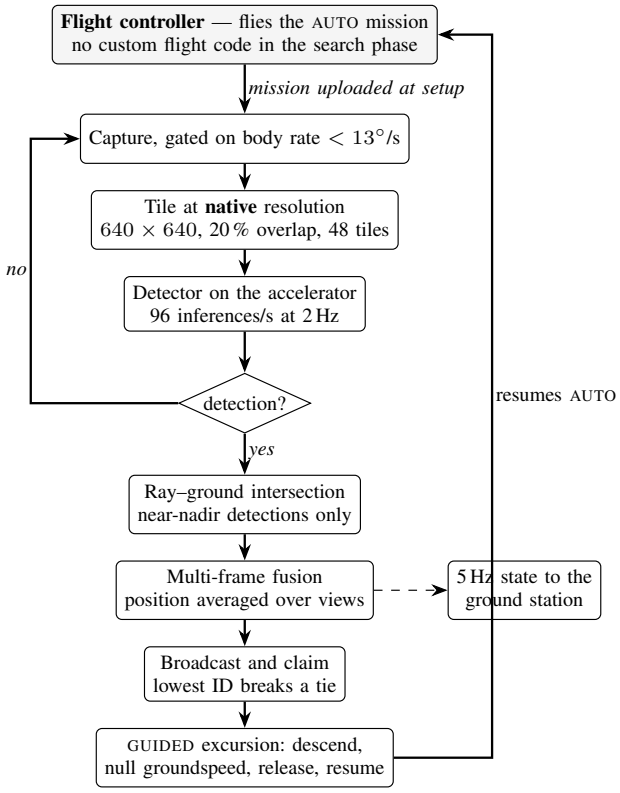


Fig. 2. Onboard processing. Detection runs alongside the sweep and never holds it up. Frames are not captured in turns, where motion blur is worst: one pixel spans 13.63 mdeg, so at a $1/1000$ s exposure a rotation of 13.6°/s blurs by one pixel, and the gate is set just below that.

When an aircraft confirms a survivor it broadcasts a claim. If two aircraft claim the same survivor, the one with the lower identifier keeps it. Abnormal conditions all end the same way: loss of link for more than 60 s, low battery and an operator abort each lead to return-to-launch. Below a second, lower battery threshold the aircraft lands where it is.

C. Onboard processing

The autopilot flies the search as an ordinary AUTO mission and would complete it with the companion computer switched off (Fig. 2). The companion detects, locates and claims survivors, and takes control only for a delivery: a short GUIDED manoeuvre to descend, stop and release, after which the AUTO mission resumes. A crash of the companion therefore costs detections, not the aircraft.

IV. Aircraft Sizing and Energy

A. Power and mass

We sized the aircraft with momentum theory [13], using cautious constants: air density of 1.150 kg/m^3 for a hot day at 300 m, a figure of merit of 0.60, motor and controller

TABLE I
FLIGHT PACK CANDIDATES, EACH EVALUATED AT ITS OWN MASS

Pack	kg	Wh	T/W	Endur.	Queue, Wh
Li-ion 6S3P	1.45	292	2.00	2.02	+1.1
LiPo 12 Ah	1.42	266	2.01	1.85*	−3.7*
LiPo 13 Ah	1.52	289	1.98	1.96*	−0.5*
LiPo 14 Ah	1.64	311	1.94	2.06	+2.5
LiPo 16 Ah HV	**1.88–2.21**	**365**	**1.8–1.9**	**2.1–2.3**	**+5 to +9**

Endurance is hover time divided by mission time and must be at least 2.0. Queue is what remains of the 20 % reserve after the third aircraft waits 120 s. * Fails. The maker does not publish the 16 Ah pack's mass, so it is bracketed at 165–194 Wh/kg.

efficiencies of 0.82 and 0.95, and 55 W of avionics. Four 18 in rotors give 0.657 m^2 of disk area. Hover shaft power is

$$P_{\text{shaft}} = \frac{T^{3/2}}{\mathrm{FM}\sqrt{2\rho A}} = 668 \text{ W} \tag{1}$$

at a take-off mass of 6.36 kg, or 913 W of electrical power once drivetrain losses and avionics are added. That is 42.2 A at 21.6 V. At the design thrust-to-weight ratio of 2.0 the electrical power rises to 2 481 W and the current to 114.8 A. Hover takes 1.59 kgf per motor, half the 3.18 kgf each motor must produce at full thrust.

The motor we selected is rated for 26.5 A continuous. It draws 10.6 A in hover and 28.7 A at full thrust, 108 % of its rating. We accept that only because full thrust is a gust reserve lasting seconds, and it is the tightest margin in the propulsion system. Pack mass, take-off mass and mission energy depend on one another, so the model iterates until they settle. Structure and battery make up 46 % of the take-off mass, the four kits 12.6 %. The fleet of three weighs 19.08 kg, inside the 25 kg limit of the Indian drone rules [14].

B. Reserve and pack selection

The battery is sized by its reserve, not by the mission. The pack has to supply the 103.4 Wh mission, a complete second sweep (30.9 Wh) for the case where the first finds nothing, and four minutes of hover (60.8 Wh) while three aircraft queue to land. That is 195.2 Wh usable, or at least 244 Wh rated if we never discharge below 20 %.

The model was first solved for an 18-cell 6S3P Li-ion pack. We then compared ready-made LiPo packs, each evaluated at its own mass, since a heavier pack needs more hover power (Table I). A 12 or 13 Ah pack fails both the endurance and the landing-queue requirement, about 14 Ah is the smallest that passes, and we chose a 6S 16 Ah high-voltage LiPo, the smallest suitable size sold in India. It also removes the need to spot-weld eighteen cells, the most hazardous step of the original build. The cost is thrust-to-weight: at 6.8–7.1 kg the motors give 1.8–1.9 unless each produces at least 3.55 kgf.

C. Voltage sag and the failsafe

A static resistance figure is enough to size a pack but not to set a failsafe, because it says nothing about when in the flight the voltage is lowest or how long a dip lasts. We simulated the chosen pack with a second-order Thevenin cell model,

$$v = \mathrm{OCV}(z) - iR_0 - v_1 - v_2, \qquad \dot{v}_k = -\frac{v_k}{R_k C_k} + \frac{i}{C_k}, \tag{2}$$

with time constants of about 5 s and 90 s and a constant-power load, so that current rises as voltage falls. The load is the mission profile scaled to the heaviest likely mass of 7.1 kg, with two two-second full-thrust gust recoveries added at 300 s and 430 s (Fig. 3).

The terminal voltage falls to 20.37 V during the late gust, below the 21.6 V low-battery threshold, at a point where the pack is still two-thirds full. A failsafe that compared raw voltage would send a healthy aircraft home. ArduPilot avoids this in two ways. It estimates the pack's internal resistance as it flies and compares a voltage corrected for the present current, and it acts only after the voltage has stayed low for 10 s. In our run the corrected voltage stays at or above 22.97 V. The low-battery action is in any case set on consumed capacity, at 20 % remaining, with voltage as a backstop. The simulation also gives a peak current of 144 A, close to the 150 A rating of the fuse and connectors, which makes the fuse the tightest electrical margin on the aircraft.

V. GEOLOCATION ERROR

A. Error budget

A survivor coordinate comes from projecting a pixel through the camera model onto a ground plane, using the aircraft's position and attitude. We modelled the total error as the root sum of squares (RSS) of independent terms and checked it by Monte Carlo simulation with 200 000 samples per case, treating each term as a two-axis Gaussian. The simulation agrees with the analytic circular Gaussian result, CEP50 = $0.8326 \times$RSS, to within one per cent.

With RTK the receiver contributes about 1 cm, which is negligible. Two terms that no receiver can reduce dominate: unmodelled error, set at 0.70 m, and the size and centring of the target, 0.50 m (Fig. 4). Together they account for 95 % of the variance. The remaining terms are boresight residual (0.16 m), the GNSS-to-camera lever arm (0.10 m), attitude (0.07 m) and pixel centroiding (0.02 m). Excluding the receiver, the budget adds to 0.88 m.

Combining views of the same target helps, but not by $1/\sqrt{n}$. At 40 m the footprint is 31.4 m long, so at 8 m/s a target stays in view for 3.9 s and is seen about eight times at a 2 Hz detection rate. Averaging reduces the random terms but not boresight error or target size, so the curve flattens by about five views. A single-frame fix without RTK gives 3.09 m RSS, and RTK with a single frame about 1.3 m. RTK with multi-frame fusion and a calibrated ground plane gives 0.91 m, or 0.94 m when attitude comes from the compass instead of a second receiver.

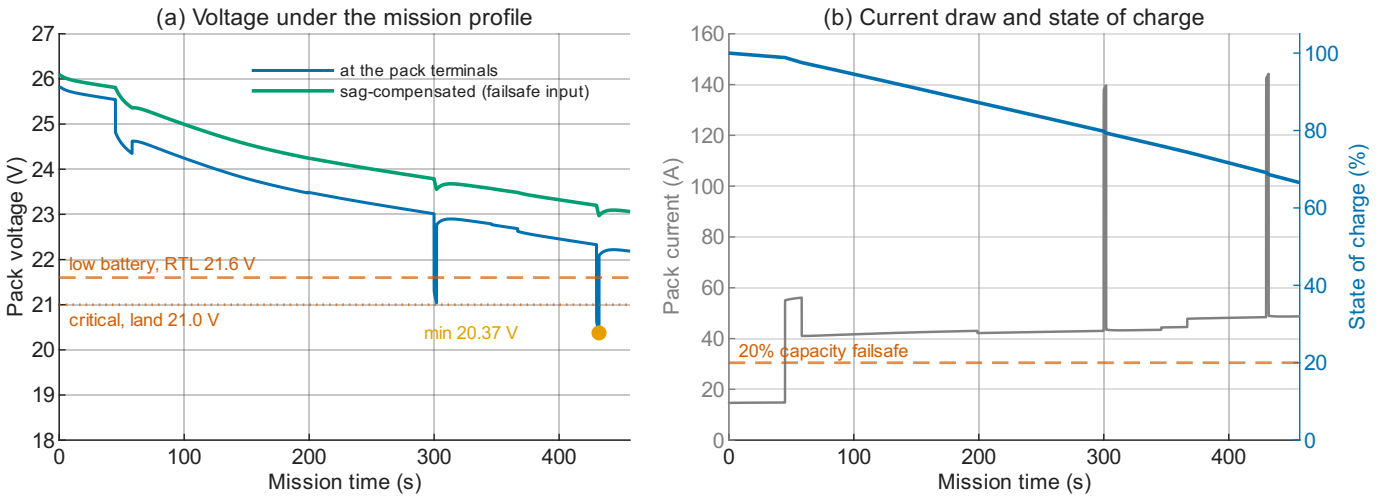


Fig. 3. Simulated voltage of the 6S 16 Ah LiPo over the mission at 7.1 kg, with two full-thrust gust recoveries. (a) Terminal voltage dips to 20.37 V, below the 21.6 V low-battery threshold, but the sag-compensated voltage the failsafe compares never falls below 22.97 V. (b) Current peaks at 144 A and the aircraft lands at 66 % charge. Cell resistances are typical values, not measurements.



Fig. 4. (a) Sources of geolocation error with RTK. Unmodelled error and target centring account for 95 % of the variance. (b) Radial error by receiver class, 200 000 Monte Carlo samples. RTK and SBAS differ by only 0.16 m in CEP50.

B. Required receiver accuracy

Adding a receiver error σ_g in quadrature to the 0.88 m floor and converting to the 95 % radius gives the requirement directly:

$$\mathrm{CEP95} = 1.7308\sqrt{\sigma_{\mathrm{rest}}^2 + \sigma_g^2} \;\le\; 5\,\mathrm{m} \;\Longrightarrow\; \sigma_g \le 2.75\,\mathrm{m}. \tag{3}$$

Table II and Fig. 5 apply this to five classes of receiver. RTK, SBAS and autonomous multi-band receivers all clear the requirement, and GPS L1 alone is marginal at 4.59 m. NavIC at its published standalone accuracy gives 8.79 m and fails. NavIC is still useful as an extra constellation for satellite geometry, but it cannot replace the correction.

The SBAS value of 0.60 m is optimistic, and field results are nearer 1.0–1.5 m. That still gives 2.31–3.01 m, so the conclusion holds. Since the mission requirement does not need RTK, we keep it for a different reason. Measuring detection recall and delivery error in field trials requires the geometry between targets, aircraft and pad to be known to centimetres, and that is what RTK provides. A ZED-F9P rover with antenna and compass now costs about as much as the receiver and

TABLE II
RECEIVER CLASSES AGAINST A 5 M DELIVERY REQUIREMENT

Class	σ_g	CEP50	CEP95	
RTK fixed	0.01 m	0.74 m	**1.53 m**	clears
SBAS / GAGAN	0.60 m	0.89 m	**1.85 m**	clears
Multi-band autonomous	1.00 m	1.11 m	**2.31 m**	clears
GPS L1 only	2.50 m	2.21 m	**4.59 m**	marginal
NavIC SPS	5.00 m	4.23 m	**8.79 m**	*fails*

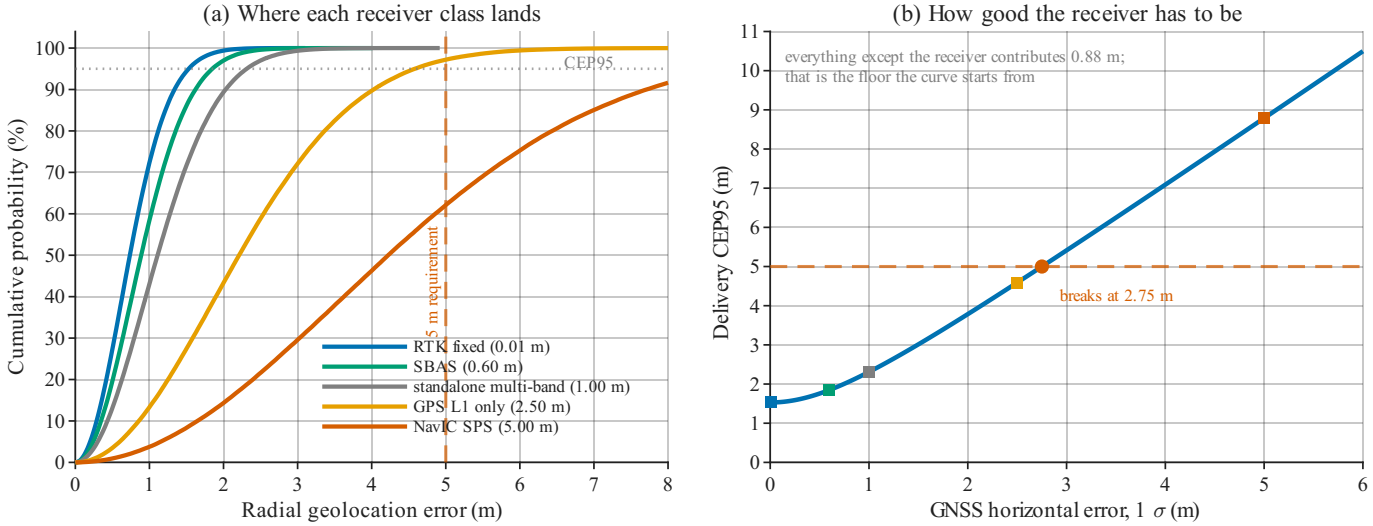


Fig. 5. (a) Radial error distributions for five receiver classes, 200 000 samples each. (b) Delivery CEP95 against the receiver's horizontal error, with the 5 m requirement and the 2.75 m limit of (3). The curve starts at 0.88 m because that part of the budget is independent of the receiver.

TABLE III
PERSON-IN-WATER TARGET SIZE BY ALTITUDE AND TILING

Altitude	Tiling	px	px^2	COCO class
40 m	**native**	**39**	**1 498**	medium
60 m	native	26	666	small
40 m	2× downsample	19	375	small
60 m	2× downsample	13	166	small

separate compass it replaces, so every aircraft carries one.

The base takes its position from its first three-dimensional fix rather than from a survey-in. This saves many minutes of setup but shifts every reported coordinate by the same absolute error, around a metre. The relative accuracy between aircraft is unaffected. A rescue crew navigating with its own metre-class receiver will barely notice the shift, and twenty minutes of survey-in would come straight out of the flying time.

VI. Detection Geometry

The camera has 4056×3040 pixels at 1.55 µm behind a 6 mm lens, giving fields of view of $55.3° \times 42.9°$ and a ground sample distance of 1.03 cm/px at 40 m. We design for the hardest target, a person in water with only head and shoulders showing, about 0.4 m across. An adult lying on a roof is about four times larger in the image.

The important quantity is the size class the target falls into, because detector accuracy is reported by size class and drops sharply for small objects. Table III gives the person-in-water target for four combinations of altitude and tiling. Only one of them, 40 m with the full-resolution frame cut into 48 overlapping 640×640 tiles, keeps the target above the COCO small-object threshold (Fig. 6). Flying higher or halving the resolution on its own is enough to push it below. At 50 m the target is already down to 959 px^2.

Full-resolution tiling has a cost in computing. At 2 Hz, 48 tiles per frame means 96 detector runs per second, against 130–160 per second published for this accelerator with the same model and input size. The 2 Hz rate gives 7.9 views per target, short of the twelve we would like for confirmation, and reaching twelve would need 3.06 Hz and 147 runs per second, at or beyond the accelerator's limit. A cheap first stage that rejects tiles of open water would free that capacity, which makes the case for a two-stage detector one of throughput rather than accuracy. We have not yet benchmarked our own model and tiling on this hardware, so these throughput numbers are the least certain in the paper.

The rest of the optics is unproblematic. The hyperfocal distance is 4.15 m, so a fixed-focus lens is sharp at every altitude flown. Keeping motion blur under a pixel at 8 m/s needs an exposure of 1/774 s or shorter, and we require 1/1000 s. The rolling-shutter readout of 25 ms skews the frame by 0.20 m, which does not affect detection but must be corrected in geolocation.

VII. Radio Links

A. From one mesh to three radios

Our first design carried video, telemetry and coordination on one 5.8 GHz 802.11 mesh. Applying the link margin

$$M = P_{\mathrm{tx}} + G_{\mathrm{tx}} + G_{\mathrm{rx}} - \mathrm{FSPL}(f, d) - S_{\mathrm{rx}} \qquad (4)$$

at the 600 m geofence, with 20 dBm into 5 dBi antennas and an MCS0 sensitivity of −82 dBm, left 8.7 dB. The link carrying the most traffic would have been the weakest. Relaying through another aircraft does not help much, because the things that break the direct path, such as a banked airframe, a person in the way or rain, tend to affect both hops.

We replaced the mesh with three radios, each carrying one kind of traffic (Table IV). Digital video runs on wfb-ng [17] at 5.8 GHz, one channel per aircraft, with a 14 dBi

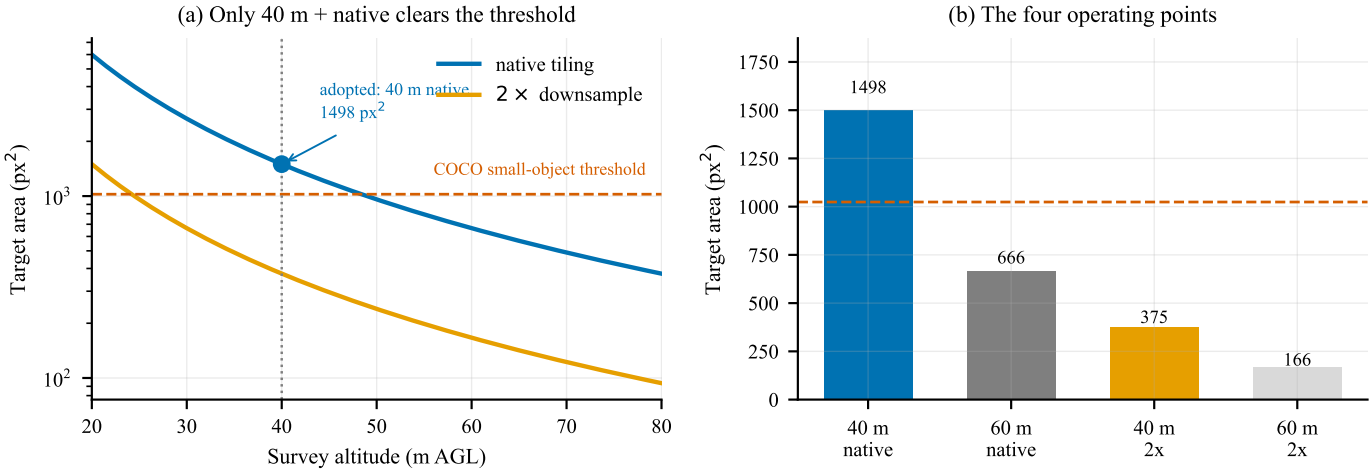


Fig. 6. Target area against survey altitude for a person in water. (a) Only 40 m with full-resolution tiling stays above the COCO small-object threshold; halving the resolution puts the target below it at every altitude. (b) The four candidate settings.

TABLE IV
RADIO LINKS AT THE 600 M GEOFENCE

Link	P_{tx} (dBm)	ΣG (dBi)	S_{rx} (dBm)	FSPL (dB)	M (dB)
Digital video, 5.8 GHz	27.0	20	−86	103.3	**29.7**
ExpressLRS, 2.4 GHz	20.0	4	−108	95.6	**36.4**
LoRa, 865 MHz SF7	14.0	4	−123	86.7	**54.3**
802.11 mesh, 5.8 GHz	20.0	10	−82	103.3	*8.7*

panel antenna at the ground for each. Command and abort run on ExpressLRS at 2.4 GHz, which also carries MAVLink telemetry [18]. Mission data and coordination run on an SX1262 at 865 MHz, which is licence-free in India. Every link now has at least 29.7 dB of margin at the geofence (Fig. 7). The cost is that one aircraft can no longer relay for another, a case the link-loss failsafe already handles.

B. Sizing the coordination channel

Splitting the links had a consequence the margin table hides. The mesh offered about 235 kbps of non-video capacity per aircraft. LoRa at SF7 on a 125 kHz channel offers 5 469 bps for the whole fleet. What limits the channel, though, is airtime rather than bit rate, because for short packets the preamble and header dominate. A 36-byte frame at SF7 takes 77.1 ms on air, where its payload alone would take 52.7 ms.

Sending state, coordination and detections as three separate messages per aircraft per second used 28.4 % of the channel, above the 25 % we allow for retries. Packing them into one 36-byte frame brought this to 25.3 %, with nothing to spare. The 54.3 dB margin pointed to the answer: the spreading factor was buying range we did not need. The same SX1262 can run GFSK at 50 kbps with a sensitivity of −104 dBm. That gives up 19 dB of margin, leaving 35.3 dB, and cuts each frame to 7.8 ms and channel use to 2.3 % (Table V). We use GFSK

TABLE V
COORDINATION CHANNEL, THREE AIRCRAFT, ONE 36 B FRAME EACH AT 1 HZ

Mode	Rate	Frame	Occupancy	Margin
GFSK 50 kbps	50 kbps	7.8 ms	**2.3 %**	35.3 dB
LoRa SF7, 125 kHz	5.5 kbps	84.2 ms	25.3 %	54.3 dB

normally and fall back to LoRa SF7 at a lower message rate when the link weakens.

The links are arranged to fail in order of importance. Video goes first and costs the operator a picture but not the aircraft a decision. Coordination degrades next, from GFSK to LoRa and then to a slower message rate. Command and abort, with 36.4 dB of margin, go last. All of these margins assume free space. Reflections off water, a low ground antenna and airframe shadowing in turns are not modelled, which is why we plan to measure packet loss against distance on the built aircraft.

VIII. PAYLOAD RELEASE

A 200 g kit tumbling with a drag coefficient of 1.05 over 117 cm² has a terminal velocity of 16.2 m/s. Integrating the fall with quadratic drag from a 6 m release gives 1.15 s to impact at 9.7 m/s, and crosswind drift of 3.24 m in a 3 m/s wind. Release velocity matters far more than release height. A residual ground speed of 0.5 m/s moves the impact point by about 0.53 m, whereas a whole metre of height error moves it by less than 4 cm. We therefore release only once ground speed is below 0.3 m/s, which keeps the fall's own contribution to about 0.32 m. Combined with geolocation and hover error, the expected delivery error with RTK is about 1.5 m RSS.

Release height over floodwater cannot come from the barometer, which measures height above the launch point. A downward lidar with a range of about 7 m measures height above the surface during the final descent and triggers the release.

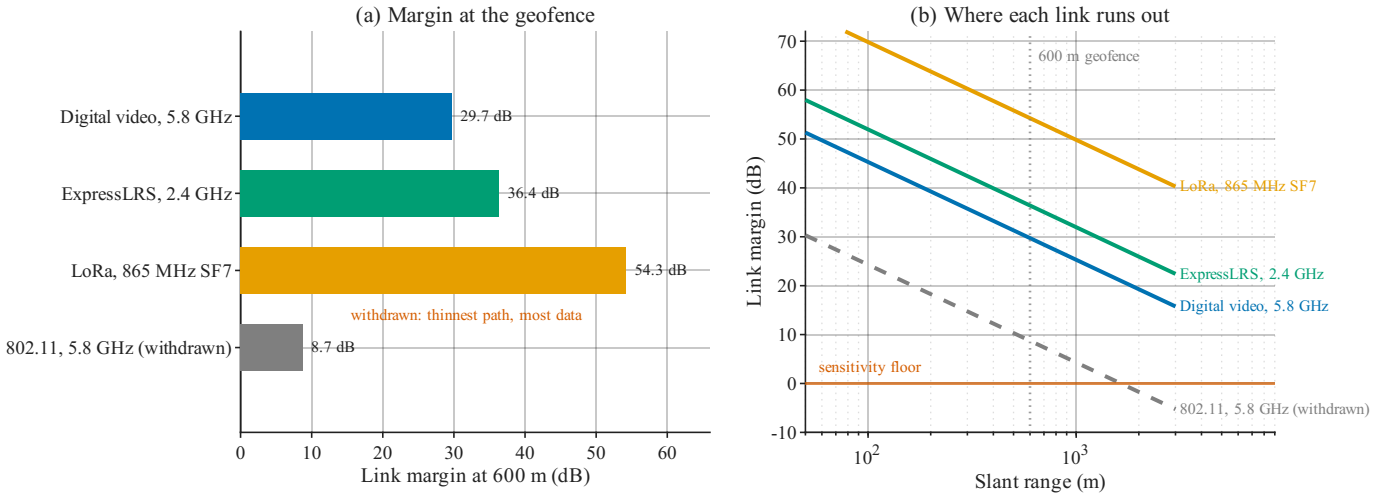


Fig. 7. (a) Link margin at the 600 m geofence, with the abandoned 802.11 mesh for comparison. (b) Margin against distance. The mesh reaches its sensitivity limit at 1.64 km; the video link would reach 18.4 km.

IX. Simulation Experiments

A. Setup

We ran three ArduCopter software-in-the-loop autopilots together, each flying its own mission file over the same search area, at real-time speed. A single telemetry router delivered all three to one ground station. The recorder logs one row per aircraft per poll. Its timestamps step backwards in a few places, so we pair aircraft by sample index rather than by time when computing separation. Separation is the three-dimensional distance between airborne aircraft, taken as those more than 1 m above the ground. Recovery covers samples where either aircraft is within 60 m of the pad.

B. Results

Launch. With all three aircraft launched together, two came within 1.31 m of each other, closer than one airframe width of 1.046 m. With the take-off delays and transit altitudes written into the mission files, the closest airborne pair during launch was 26.96 m apart (Fig. 8). The aircraft left the pad at 0.0, 15.5 and 31.7 s against commanded delays of 0, 15 and 30 s.

Sweep. Away from the pad the aircraft never came closer than 35.16 m. Choosing the sweep direction to finish near home brought all three sweeps to an end 117–130 m from the pad (Fig. 9). Under our earlier rule, which ignored the end point, two aircraft finished 516 m and 540 m away, on the lowest battery of the flight. The path length is the same either way.

Recovery. With the landing slots in a row the slot spacing was 1.22 m, less than an airframe width. Placing them at the corners of the pad's usable square raised it to 2.61 m at no cost (Fig. 10). The closest airborne approach during recovery was 5.34 m. The corner layout has one weakness: each slot is exactly one airframe radius, 0.523 m, from the pad edge. A stationary aircraft on its slot fits, but any touchdown error puts part of the airframe over the edge. Moving the slots inward would reduce the spacing that the corners were chosen for, so the layout can only be settled once touchdown error has been measured in flight.

Failsafe and ground station. In a separate run the low-battery failsafe returned each of the three aircraft to the pad, triggering at 10 809 mAh consumed against a configured 10 800 mAh. The ground station displayed all three aircraft at once from offline map tiles (Fig. 11).

X. Discussion

Several of our results say more about the design process than the aircraft. The receiver analysis is the clearest example. We expected RTK to be the deciding factor in coordinate accuracy, and it turned out to buy 0.16 m of CEP50 over SBAS, because target size and unmodelled error dominate. The link analysis is another. The mesh was attractive for its relay capability, and the margin calculation, not the architecture diagram, showed that it would have been the weakest link in the system. In both cases a few lines of arithmetic changed a decision that looked settled.

We also learned something about verification. Four times during development we found a configuration file that was correct and reviewed but never loaded by the running system. In one case the failsafe parameters were never applied in simulation, so every test had run with the low-battery action switched off. Since then we count a behaviour as shown only once the real configuration has run end to end and its values have been read back from the running autopilot.

A. Limitations

Nothing has flown, so none of the physical effects is measured. The most important unknown is detection recall on real flood imagery. We do not train on images of real casualties, and our field targets will be home-made forms laid on ground surveyed with our own RTK rover. Because the reference positions come from the same kind of receiver as the aircraft, their errors are correlated, and measured geolocation

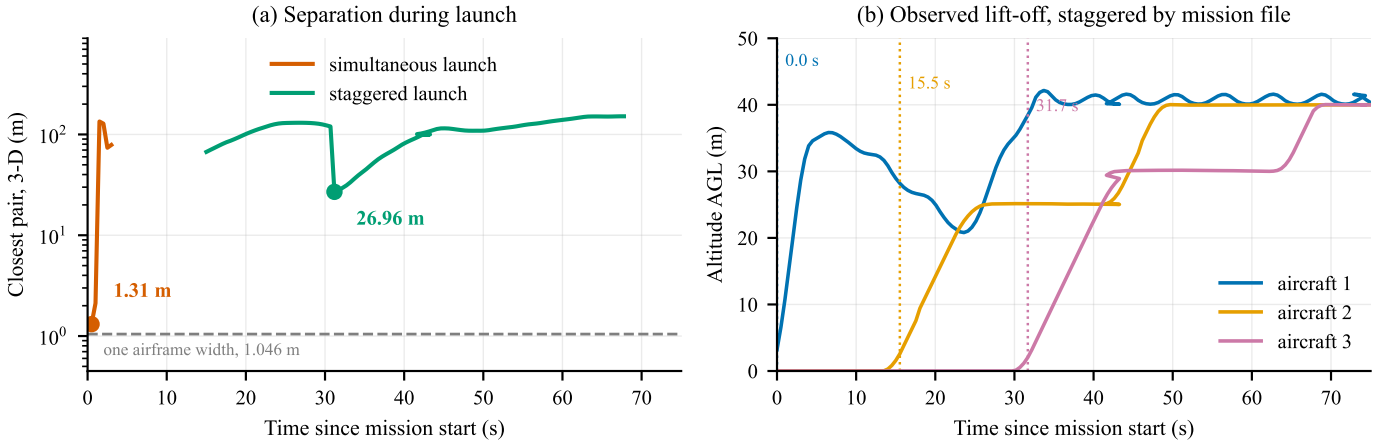


Fig. 8. Separation at launch from the recorded telemetry of three ArduCopter autopilots. (a) Closest airborne pair in three dimensions, log scale. Simultaneous launch brings two aircraft within 1.31 m; staggered take-off keeps them at least 26.96 m apart. (b) Altitude of each aircraft, with observed lift-off times.

accuracy will look slightly better than it would against an independent survey.

The motors publish no thrust curve, so the thrust-to-weight ratio is a requirement until it is measured on a thrust stand. The battery model uses typical cell parameters rather than measured ones. The accelerator throughput figures are published values, not our own.

The system cannot see obstacles ahead. A 40 m survey altitude clears the two- to four-storey buildings of rural and semi-urban flood plains but not those of Chennai or Mumbai, and flying higher is not an option because at 50 m the target already drops below the small-object threshold. Tall structures must be drawn as exclusion zones, which the autopilot enforces, and an unmarked one would be hit. The system is therefore suited to open water and low-rise flooding only.

B. Data protection

The system records the locations and images of people in distress who cannot consent. Under India's Digital Personal Data Protection Act 2023 this is personal data. Coordinates are kept only for the operation and deleted after handover, images stay on the aircraft unless needed as evidence for a specific measurement, and no image of an identifiable person is published.

XI. Conclusion and Future Work

We have presented the design of a three-aircraft system for flood search and relief delivery and evaluated it by analysis and simulation. Coordinate accuracy is limited mainly by target size and unmodelled error, so any receiver better than 2.75 m meets a 5 m requirement. Detecting a person in water needs 40 m and full-resolution tiling together. Separate single-purpose radios give every link at least 29.7 dB of margin, and framing overhead rather than bit rate sets the capacity of the coordination channel. In simulation, scheduling take-off in the mission file raised launch separation from 1.31 m to 26.96 m.

The next step is to build and fly the aircraft and replace the modelled figures with measured ones, starting with thrust, pack sag, detection recall and delivery error. Two extensions interest us beyond that. For night operation over water, near-infrared imaging may suit better than thermal: a person in floodwater cools towards the water's temperature, while water absorbs strongly above about 900 nm and skin does not. And running the same tiled detection on accelerators of decreasing capability would show how much computing a given coverage rate really needs, which we have not found published for aerial person detection.

Acknowledgment

We thank Dr. Surbhi Sharma and Dr. Mamata Gulati for their supervision and guidance, and the Department of Electronics and Communication Engineering, Thapar Institute of Engineering and Technology, for its support.

References

[1] Internal Displacement Monitoring Centre, *Global Report on Internal Displacement 2025*, Geneva, 2025.

[2] A. C. Kapoutsis, S. A. Chatzichristofis, and E. B. Kosmatopoulos, "DARP: Divide areas algorithm for optimal multi-robot coverage path planning," *J. Intell. Robot. Syst.*, vol. 86, pp. 663–680, 2017.

[3] H. Choset, "Coverage of known spaces: the boustrophedon cellular decomposition," *Auton. Robots*, vol. 9, no. 3, pp. 247–253, 2000.

[4] H.-L. Choi, L. Brunet, and J. P. How, "Consensus-based decentralized auctions for robust task allocation," *IEEE Trans. Robot.*, vol. 25, no. 4, pp. 912–926, 2009.

[5] T.-Y. Lin *et al.*, "Microsoft COCO: Common objects in context," in *Proc. ECCV*, 2014, pp. 740–755.

[6] X. Yu, Y. Gong, N. Jiang, Q. Ye, and Z. Han, "Scale match for tiny person detection," in *Proc. IEEE/CVF WACV*, 2020.

[7] F. C. Akyon, S. O. Altinuc, and A. Temizel, "Slicing aided hyper inference and fine-tuning for small object detection," in *Proc. IEEE ICIP*, 2022, pp. 966–970.

[8] S. Sambolek and M. Ivasic-Kos, "Automatic person detection in search and rescue operations using deep CNN detectors," *IEEE Access*, vol. 9, pp. 37 905–37 922, 2021.

[9] P. Zhu *et al.*, "Detection and tracking meet drones challenge," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7380–7399, 2021.

[10] L. A. Varga, B. Kiefer, M. Messmer, and A. Zell, "SeaDronesSee: A maritime benchmark for detecting humans in open water," in *Proc. IEEE/CVF WACV*, 2022.

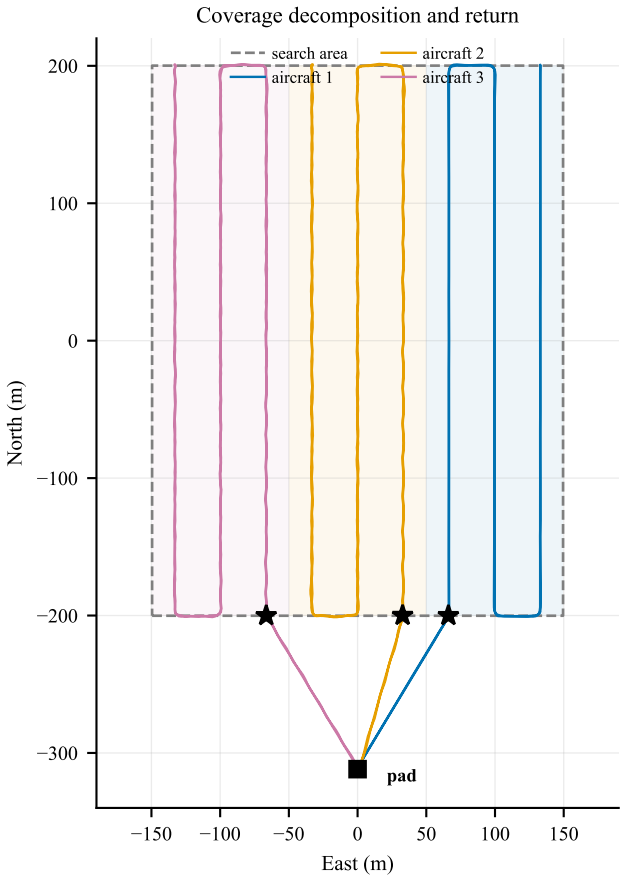
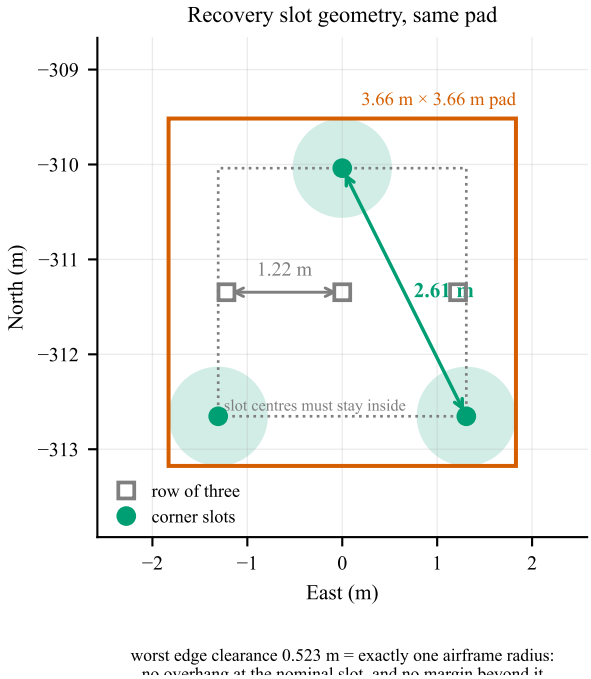


Fig. 9. Equal-area strips and recorded tracks. Stars mark the end of each sweep, all within 117–130 m of the pad.



worst edge clearance 0.523 m = exactly one airframe radius: no overhang at the nominal slot, and no margin beyond it

Fig. 10. Landing slot layout on a 3.66 m pad. Slot centres must stay half an airframe inside the edge. A row gives 1.22 m spacing and the corners give 2.61 m. Discs show the 1.046 m airframe footprint.

[11] S. G. Mathisen, F. S. Leira, H. H. Helgesen, K. Gryte, and T. A. Johansen, "Autonomous ballistic airdrop of objects from a small fixed-wing unmanned aerial vehicle," *Auton. Robots*, vol. 44, pp. 859–875, 2020.

[12] T. Takasu and A. Yasuda, "Development of the low-cost RTK-GPS receiver with an open source program package RTKLIB," in *Proc. Int. Symp. GPS/GNSS*, 2009.

[13] J. G. Leishman, *Principles of Helicopter Aerodynamics*, 2nd ed. Cambridge, U.K.: Cambridge Univ. Press, 2006.

[14] Ministry of Civil Aviation, Government of India, *The Drone Rules, 2021*, G.S.R. 589(E), Aug. 2021.

[15] ArduPilot Development Team, "ArduPilot." [Online]. Available: https://ardupilot.org/

[16] u-blox, "ZED-F9P high precision GNSS module," data sheet, 2024.

[17] wfb-ng contributors, "wfb-ng: long-range packet radio link based on raw WiFi." [Online]. Available: https://github.com/svpcom/wfb-ng

[18] ExpressLRS Project, "MAVLink," ExpressLRS documentation. [Online]. Available: https://www.expresslrs.org/software/mavlink/

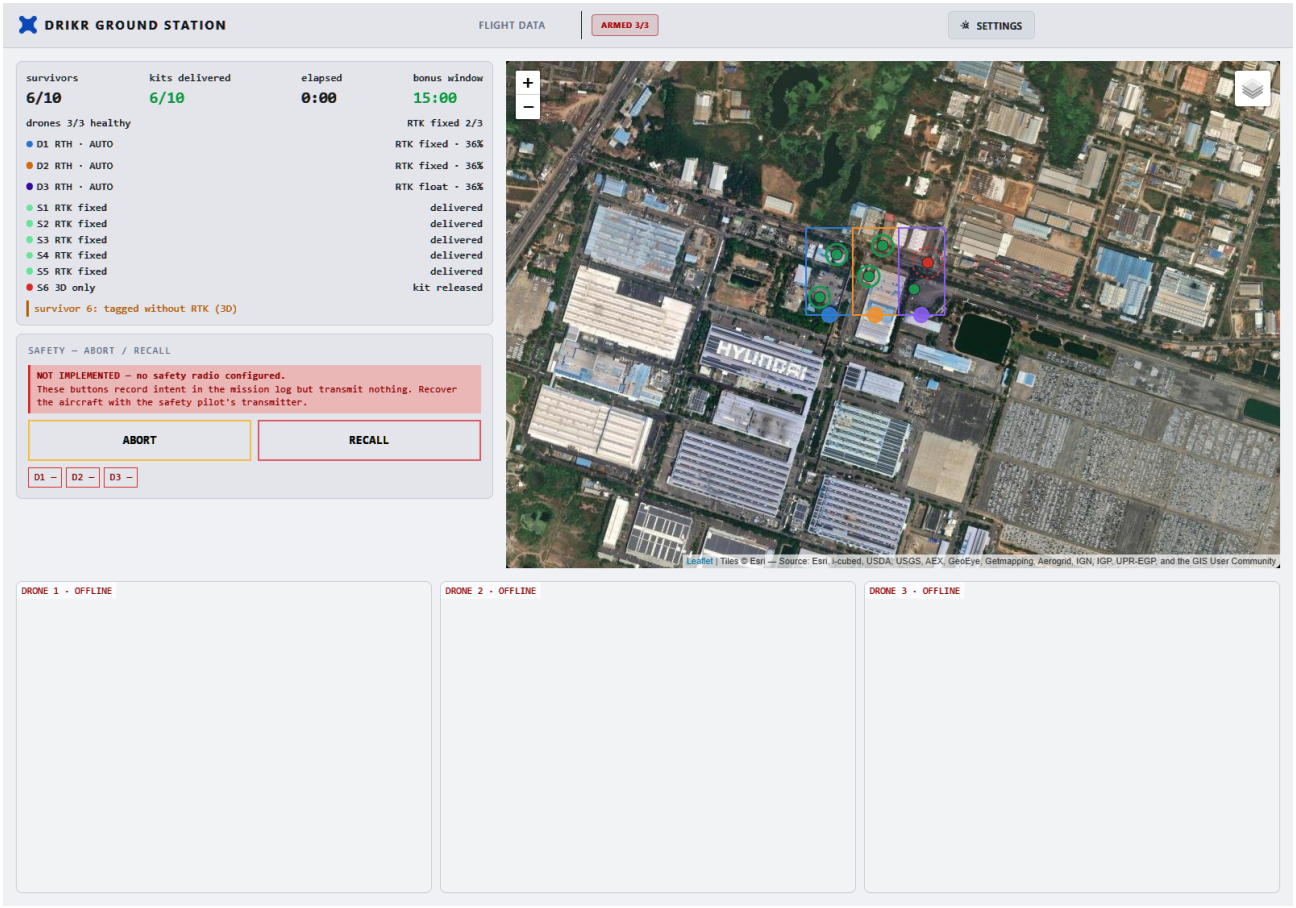


Fig. 11. Ground station during a simulated mission. The interface runs with a dark theme; for print, the panel colours have been inverted to a light scheme with their hues kept, and the map is shown as captured. Nothing else has been changed. The telemetry comes from a test driver, not from aircraft. Six survivors have been found and six kits dropped. Five survivors are tagged RTK FIXED and the sixth 3D ONLY; the display shows the difference because a crew sent to the sixth must search a wider area. The safety panel states that abort and recall are not yet transmitted. The video panels are offline because the media server was not running.